PROCEEDINGS

Two Days National Conference on

ARTIFICIAL INTELLIGENCE FOR DECISION MAKING

10th & 11th March 2023

Dr.L.M.R.J.Lobo

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

IN ASSOCIATION WITH

P.A.H.SOLAPUR UNIVERSITY, SOLAPUR

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Shriman Seth Arvindji Doshi

Eminent Industrialist
Chairman, S.A.P.D. Jain Pathshala, Solapur.

Message

The new industry revolution 4.0 forced Indian industries to invest in research and innovation. The technology advancement is growing rapidly and I believe that it will remove all the obstacles in the development of our country.

I hope the Two day National level Conference on Artificial Intelligence for Decision making on 10th and 11th March 2023 organized by Walchand Institute of Technology, Solapur in association with P.A.H. Solapur University, Solapur will provide an opportunity to all the participants to interact with each other and brainstorm on the current challenges in all sectors.

I am sure that this conference would prove to be outstanding, one amongst the many conferences being conducted on this theme.

I congratulate Dr. Vijay Athavale and his team for their commendable work in organizing AIDM 2023 and wish the conference a grand success.

Shriman Seth Arvindji Doshi

Dr.Ranjeet H. Gandhi

Hon. Secretary
S.A.P.D. Jain Pathshala, Solapur

Message

In today's speedy dynamic world, there is a requirement for a forum for exchange of innovations and ideas. It is gratifying to notice that Walchand Institute of Technology, Solapur is organizing a Two days National level Conference on Artificial Intelligence for Decision making on 10th and 11th March 2023.

I am sure that the conference will inspire every curious participant and adventurous mind. They will be encouraged to dream big and work hard to solve industry oriented problems. The theme of the conference is interdisciplinary and will explore the knowledge of participants. I do hope that deliberations of the conference will be able to present a definite action plan to take the benefit of the technical revolution to society.

I am happy to note that we have received papers from pan India. I sincerely comprehend the efforts taken by the Principal, Faculty and Staff and hope all the participants will enjoy the thought provoking sessions and hospitality at W.I.T .

I wish best luck for successful conduct of the conference AIDM 2023.

Dr. Ranjeet H. Gandhi

Dr. Vijay A. Athavale

Principal
Walchand Institute of Technology, Solapur

Message

As Principal of Walchand Institute of Technology, Solapur, I extend warm and happy greetings to all the delegates and participants of the two days national level conference on Artificial Intelligence for Decision Making (AIDM 2023). The event will make researchers become aware, grow and broaden their horizon of thought process by indulging into this emerging area, Artificial Intelligence.

We are living in a time where we have to break boundaries in various dimensions and share required knowledge and advancements at a rapid pace .The conference will help to develop an awareness of AI as a stream of research for the new future.

The AIDM 2023 aims to bring different disciplines under one roof and provide opportunities to exchange ideas for building research relations for future collaboration. The theme is indicative of relevant research in AI that will give authors innovative prepositions about the ambit of discussion.

I hope AIDM 2023 will help Faculty, Research Scholars, U.G. and P.G. students to have the latest updates for better understanding of today’s engineering industry challenges and contribute to the progress of the nation.

All this would not be possible without financial support from P.A.H. Solapur University,Solapur and our management. I am thankful to all contributors, reviewers, and experts for their constant support in making the national conference AIDM 2023 more remarkable.

Dr. Vijay A. Athavale

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(Eminent Industrialist)
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AIDM 2023

THEME INDEX

Theme 1

Social

INDEX

Social_01

SOLAR BASED AUTOMATIC IRRIGATION SYSTEM

Prashant Dharashive [1] , Manish Sawale [2]

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Abstract :

IOT agriculture started with a simple idea - Removing guesswork from farming.
Farming is not easy and involves a lot of uncertainties. Farmers make most of these decisions based on their own experience, knowledge passed on from their elders or recommendations from their fellow farmers. However, our farmers have found themselves in a scenario where their ways of decision making have remained the same but everything around it has changed including the climate, the soil, the agro chemicals, the seeds, the fertilizers, etc. IOT agriculture felt that a massive disruption was needed in the way our farmers are making farm level decisions. These decisions cannot have any element of guesswork.
IOT agriculture saw that these uncertainties pondered the farmer's mind not just during specific crop stages, but every minute; every hour and every day while the crop was standing in their farms. IOT agriculture realized that only data driven farming would take the guesswork out of farming by pinpointing exactly what works and also what doesn't work.

Keywords: Water Management, Irrigation System, Surface Drip Irrigation, Solar Based Irrigation System, Automation Based Irrigation System.

Social_02

PWA-Enabled Mediapipe-based Fitness Training: Improving Exercise and Yoga Form and Performance in Real-Time

Amarja Agaonkar [1] , Angad Singh Obbi [2] , Atharva Mulgund [3], Tejas Phanse [4]

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Building of the progressive website in an attempt to give users the experience of a gym trainer in the comfort and flexibility of home using the Media pipe library which tracks the body movements that match the exercise form to improve mind-muscle connections. The aim is to detect a person's 33 body joints from a perfect view and measure from neck to torso inclination to some reference axis. By monitoring the inclination angle when the person bends below a certain threshold angle and calculating reps. Other features include measuring the time of a particular posture and the camera alignment (yoga pose). We need to make sure that the camera looks at the proper view. Hence the alignment feature is required and integrating this functionality on a website provides a pattern for the tasks to be carried out and the diet to keep you healthy.

Keywords: React, Material UI, PWA Media Pipe, COCO model, Computer visions, Body joints.

Social_03

Real-Time Face Recognition for Biometric Attendance Using Convolutional Neural Network

Pooja Shivkumar Katake [1], Rupali J. Shelke [2]

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Abstract :

Face Recognition is a computer program that can find, follow, recognize, or confirm human faces in photos or videos taken with a digital camera. An attendance monitoring system using deep learning techniques would capitalize on the promise of this technology in the area of face detection and identification, replacing the current attendance system, which has always been time-consuming and problematic for staff of institutions and organizations. Take a picture of the students in the classroom first, then use the OpenCV module to identify and frame their faces. Build a convolution neural network (CNN) as the project's final step to train these facial photos, compare them to student records in the database, and update the status of the students' attendance. Our system guarantees a hassle- free attendance system that is simple to maintain, and with additional connectors, it may be used for other demands of businesses. limited to educational institutions.

Keywords: CNN, Attendance, Face Recognition.

Social_04

Face Recognition Attendance System

Kedar Sathe[1], Sunita M Dol[2] , Sahil Manikshete[3], Vallabh Kulkarni[4], Rounak Sancheti [5]

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Abstract :

Student Attendance mainframe structure is defined to manage the student's class attending files using the concept of face detection and recognition through open computer vision. The principal reason this system has been put forward is to improve the traditional attendance system of various universities to avoid the misuse of time and assets. The pointing-sides of the automation world have forced an idea of switching from standard attendance to the digital system by using face detection and recognition methods. The major reason for building this system is to improve the adaptability and performance of the attendance system procedure besides reducing the long-term time load, work and disposables used. The main purpose of the Student Attendance markup structure is to perform, adding and manipulating attendance notes of an individual, automatic calculation on number of present and absent based on subject and affability of the class and then generates the automated document or spreadsheet. This idea is completely based on a general purpose language named python through which we use the concept of open computer vision. For the face detection system we used Haar-cascade and for face recognition, we used the LBPH model; then the training of individual students happened and finally the system generates the spreadsheet which provides the no. of students present in the classroom with an image or video capturing live..

Keywords : Local Binary Patterns Histograms (LBPH), Python, Face Detection, Face Recognition, smart attendance, Haar - Cascade, OpenCV, Tkinter

Social_05

Implementation of FingerPrint Voting System Using Machine LearningAlgorithm

Prince Singhvi [1], Shivappa M Metagar [2], Chirag Jain [3], Rahul Jain [4], Chayan Jain [5], Samyak Jain [6]

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Abstract

It has always been a very difficult task for the election commission to conduct free and fair elections in the country. Crores of rupees have been spent on this to make sure that the elections are completely free from all the malpractices. But, now- a -days it has become common for some forces to perform some malpractices which eventually leads to the unfair elections. We have designed a voting system which is based on fingerprints where there is no need for the user to carry his/her Voter ID to the polling booth. The voter just needs to keep the finger on the machine and the data is retrieved from the database if it matches then the voter is allowed to vote otherwise denied. And Thus Conducting a free from malpractices and a speedy election is carried out easily.

Keywords

Biometrics, Fingerprint Sensor, Validating users, Voting, Website, Online e-voting

Social_06

Face Emotion Recognition using CNN

Pranoti Kodam [1], L.M.R.J. Lobo [2], Sakshi Nipunage[3]

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A**bstract** :

Face Recognition is a computer program that can find, follow, recognize, or confirm human faces in photos or videos taken with a digital camera. An attendance monitoring system using deep learning techniques would capitalize on the promise of this technology in the area of face detection and identification, replacing the current attendance system, which has always been time-consuming and problematic for staff of institutions and organizations. Take a picture of the students in the classroom first, then use the OpenCV module to identify and frame their faces. Build a convolution neural network (CNN) as the project's final step to train these facial photos, compare them to student records in the database, and update the status of the students' attendance. Our system guarantees a hassle- free attendance system that is simple to maintain, and with additional connectors, it may be used for other demands of businesses. limited to educational institutions.

Keywords: CNN, Attendance, Face Recognition.

Social_07

Analysis of Water Quality using Machine Learning

Kajal Rajendra Gavali [1], A. S. Gundale [2], S. R. Gengaje [3], R. J. Shelke [4]

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Abstract :

Water is the most important thing in human daily life such as human drinks, farming and also manufacturing, different industries. However, urbanization modernization and industrialization had caused water pollution. Therefore, it is important to have a water quality monitoring tool which provides real-time water quality data. A practical and convenient water quality monitoring system is designed in this project, which is based on the MCU (Micro-programmed Control Unit) and Bluetooth technology. This design takes the Arduino board based on ATMega328P chip as the core and uses sensors to collect water parameters like pH, turbidity, conductivity and temperature. This system obtains the water quality parameters in time and accurately. The measured data after processing is used to decide the quality of water using a machine learning algorithm.

Keywords: Human,Urbanization,Modernization And Industrialization, Water Quality Monitoring

Social_08

Augmented Reality in Education

Ashwini Adsule [1], P. S. Shetgar [2], Abhilash Birajdar [3], Swapnil Adaki [4]

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Abstract :-

Virtual Things Are Superimposed Over The Real-World Surroundings In Augmented Reality. Educators Understand That Learning Is Enhanced Not Only By Reading And Hearing But Also By Creating And Participating. Our Suggested Idea Attempts To Use Augmented Reality To Improve The Current Educational System. Another Aspect Of The Ar Experience Is That It Includes 25% Digital Reality And 75% Existing Reality. It Means It Doesn't Replace The Complete Environment With The Virtual; Rather, It Integrates Virtual Objects Into The Real World.

Keywords: Augmented Reality ,Learning ,Reading , Creating , Ar Experience

Social_09

AI Driven Career Guidance System

Pranali Bavdhankar [1], Tulsi Kokare [2], Lavanya Aken [3]

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Abstract :–

Career guidance is a way of engaging students about their careers .Career guidance system provides students not to be confused about their decision about their career. The system makes students mature in deciding a right career path. It brings out students' interest to pursue their career and guides the student in the right way.But now, parents and students know the importance of choosing the right career path. Students struggle to understand the career options and opportunities open to them as they have limited access and exposure to information and opportunities. Nowadays students and parents are realizing that they want to move towards a career path in which they are interested and passionate about. But in reality, most of the students are unable to choose their passion and interest to find out the right path towards a career. Our proposed system Careers4U is an Artificial Intelligence and Machine Learning based Career guidance System for students struggling with choosing a right career path. This platform assists students to explore their interests using Howard Gardner theory of Multiple Intelligence and assists to choose future career paths and skilling options.We provide real-time insights of your top three intelligence and possible future career options waiting to welcome you. you. It is a recommendation system for students and learners in taking informed upskilling courses from DIKSHA ,Swayam and other Skill platforms worldwide to train oneself in the areas of students aspiration. It also refers students to some of the successful people and further we refer you to top colleges, Mentors and direct students for career counseling and internships.

Keywords : Career Guidance System, Multiple intelligence, Artificial Intelligence (key words)

Social_13

DeepLeaf: AI Application for Plant Disease Detection

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Abstract :

DeepLeaf is an AI based web application designed to accurately identify plant leaf diseases using deep learning algorithms. The application uses a convolutional neural network (CNN) model trained on a large dataset of plant leaf images to analyze and classify images of plant leaves with high accuracy. The user can upload an image of a plant leaf to the DeepLeaf application, and the CNN model will analyze the image and classify it according to the specific disease affecting the plant. The application also provides information on the symptoms, causes, and recommended treatments for the identified disease. DeepLeaf's CNN model was trained using Plant Images. The model was fine-tuned on a dataset of plant leaf images with 22 classes, including a wide range of plant diseases. The model achieved an accuracy of 96.34% on a test set of plant leaf images, demonstrating its high accuracy in identifying plant diseases. The DeepLeaf application is user-friendly and accessible to anyone with an internet connection. It can be used by farmers, gardeners, and other plant enthusiasts to diagnose plant diseases quickly and accurately, allowing for timely treatment and prevention of further spread of the disease.

In conclusion, DeepLeaf is a valuable tool for accurate and efficient plant disease identification. The use of deep learning algorithms, specifically CNN models, has shown to be highly effective in accurately identifying plant diseases. Further research could explore the performance of the application on a wider range of plant species and disease types, as well as the integration of additional features such as real-time disease monitoring and alerts.

Keywords : (Plant Leaf Disease Identification, Deep Learning, Artificial Intelligence, Image Classification.)

Social_14

Detection of Excavator and Workers at Construction Site using YOLO V3

Shifa Patel [1] , Meenakshi N. Shrigandhi [2] , Rachana Chimman [3] , Ritika Chilka [4]

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Abstract :

Object detection is a crucial area of computer vision that has seen significant advancements with the introduction of deep learning techniques. YOLO is one of the most well- liked algorithms for object detection. This research paper proposes an approach for the detection of excavators and workers at construction sites using YOLO v3, an object detection model. The construction industry is inherently dangerous, and accidents often occur due to the presence of heavy machinery and workers in close proximity to each other. The proposed approach uses YOLO v3 to detect excavators and workers in real-time, allowing for the implementation of safety measures to reduce the risk of accidents. The effectiveness of the proposed approach was evaluated through experiments on a dataset of construction site images, and the results show high accuracy rates in detecting excavators and workers. The proposed approach can contribute to improving safety in the construction industry and reducing the risk of accidents

Keywords : YOLO v3,Heavy Machinery And Workers,Excavators, Risk Of Accidents

Social_15

HR Employee Attrition Analysis

S. N. Chamatagadour [1], Nikhita Sargam [2], Maheshwari Sindam [3], Ramkrushna Karnikanti [4]

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Abstract :

The recent increase in the technological capacity to gather large magnitudes of data and analyze it has changed the way in which decision makers use them to decide on making the optimal decision. Employee attrition very similar to customer churn is an important and deciding factor affecting the revenue and success of the company. To avoid this problem, many companies at the moment are taking guidance via machine learning strategies to expect the employee churn/attrition. In this paper we are analyzing data from past and present using different classification like Random forest (RF) to come up with a better predictive model for the dataset present. Through this we are hoping to help the company to predict employee churn and take effective measures to retain the employees and improve their economy loss due to the loss of valuable employees

Keywords : Machine Learning; Employee Attrition; Random Forest ; Decision Tree ; Feature Ranking ; Feature Selection

Social_16

Emotion Recognition Using Artificial Intelligence

Mudassar Inamdar [1], **Aditya More** [2], **Aquib Mokashi** [3] , **Avinash Kulkarni** [4],

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Abstract :

The interplay among humans and computer systems may be extra herbal if computer systems are capable of understanding and replying to human non-verbal communique consisting of feelings. Although numerous methods were proposed to understand human feelings primarily based totally on facial expressions or speech, exceedingly restrained conditions have been achieved to fuse those two, and other, modalities to enhance the accuracy and robustness of the emotion reputation system. This paper analyses the strengths and the constraints of structures primarily based totally handiest on facial expressions or acoustic information. Facial expression reputation is a part of Facial reputation that is gaining extra significance and wants for it will increase tremendously. Though there are strategies to pick out expressions using device gaining knowledge and Artificial Intelligence techniques, this technique tries to apply deep gaining knowledge and photo type approach to understand expressions and classify the expressions in line with the images.

Keywords : Facial Reputation; Expression Reputation; Deep Gaining Knowledge Of; Photo Reputation; Facial Technology; Sign Processing; Photo Type.

Theme 2

Application

INDEX

Application_01

Job Scheduler – Using Machine Learning

Aveena Dange [1], Lekhraj Vilhekar [2]

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Abstract :

CPU scheduling is the basis of multiprogrammed operating systems. By switching the CPU among processes, the operating system can make the computer more productive. This paper is based on methods aimed at using Machine Learning to schedule our processes. The research highlights the need for and importance of scheduling and the various machine learning algorithms to make predictions of which one is ideal for job scheduling. In the Machine learning job scheduler - Snakepit, we aim to understand its features, implementation, and working by studying the architecture of the system.

Keywords - Operating System, Machine Learning, job scheduler, Snakepit

Application_02

Early Detection ML Algorithm to turn down Cancer Mortality.

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Abstract :

Cancer can kill when tumors affect the function of major organs. Life threatening complications can also occur due to malnutrition, a weakened immune system, and lack of oxygen. Cancer treatments can prevent some of these complications, as well as disease progression to its next stage.Chinese scientists say a new blood test can detect five kinds of cancer up to four years before a doctor could. As we are aware of how rapidly cancerous cells divide and redivide giving rise to tumor, tests like PanSeer can reduce the cancer mortality. It is possible to detect cancer 4 years in advance, increasing the chances of survival and resisting its next stage.PanSeer is a technology developed by a team of researchers from the University of California, San Diego (UCSD) and it uses machine learning algorithms to detect cancer from blood samples.PanSeer is a novel blood test that has been developed to detect early-stage cancer by identifying small fragments of tumor DNA that circulate in the bloodstream. The test has been shown to be particularly effective in detecting cancers that are difficult to diagnose through traditional screening methods, such as pancreatic, liver, and esophageal cancers.PanSeer is an early detection test which puts together existing technologies for many different genome tests.

Keywords : Cancer, PanSeer, Tumors, ct DNA, PCR, DNA Marker.

Application_03

Machine Learning based classification of medical X-ray images and fracture

identification with convolutional neural networks

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Abstract :

A huge number of the population in underdeveloped countries are unable to access the appropriate healthcare services. In the current paper, we propose a machine learning based automation tool to identify the X-ray images segmentation and aid the medical professional in identifying the fractures. The following five structural classifications should be applied to digital X-ray images: elbow, leg, spinal cord, tows, and other things x-ray. In this paper, the outcomes of an experimental evaluation of the classification of X-ray images for the Imaging CLEF-2019 challenge are reported. Contemporary feature extraction and classification methods are utilized to categorize the images and they are optimized for the challenging task with a focus on features that show the size and shape of the bones.

Keywords: X-Ray Images, Classification, CNN, Fracture detection.

Application_04

COVID-19 Detection from Lung CT Imagery using AI

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Abstract :

COVID-19 pandemic has taken over the world from December 2019. Worldwide 414,019,118 COVID-19 cases are detected to date. Many techniques are introduced for detecting the virus, but we have taken one step ahead by finding its severity automatically. The suggested technique will assist in quickly screening out potential Coronavirus-infected individuals as well as the severity for any further diagnosis and treatment. The identification of “severe acute respiratory syndrome coronavirus 2 that causes the disease 2019 novel coronavirus (COVID-19)”, utilizes Lung Computed Tomography (CT) which is essential for both physicians and patients. Successful monitoring of infectious cases is a crucial phase throughout the battle over COVID-19, utilizing the CT scan is becoming one of the main monitoring strategies. Lung CT scans are a more accurate screening test for diagnosing lung disorders. Artificial intelligence seems to be the most effective machine learning method. This provides useful methods to analyze a vast number of CT images, which could have a significant effect on COVID-19 testing. This paper suggests a convolutional neural network that combines the Inception and Auxiliary layers to a deep dense network for the training of CT images. As a result, we use GoogLeNet, a deep convolutional neural network architecture optimized for the detection of COVID-19 case situations in CT images. The severity gives the information about the patient’s condition in four stages which are Normal, Mild, Moderate, and Severe. The algorithm has achieved a good level of accuracy (90.66%), precision (92.1%), and sensitivity (91.8%). Automatic detection of COVID-19 and its severity helps more in the treatment process. The time required for analyzing the result is less by which more lives can be saved.

Keywords : CT scan, Severity, GoogLeNet, COVID-19, Inception

Application_05

A System for Prediction and Analysis of Cancer Disease Using Machine Learning Algorithms

Prathamesh Siddharam Birajdar [1] , L.M.R.J. Lobo [2]

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Abstract :

Screening for cancer remains a very sensitive and hotly debated issue in medical practice. An analysis of published data reveals a multitude of opinions based on a limited amount of reliable data. There is continuing controversy regarding the appropriate age limits for screening mammography and in fact, concerning the value of mammography itself. Eventually, there is an ambiguity about the use of screening for lung or prostate cancer being meaningful. Recommendations and decisions regarding cancer screening should be based on reliable data and not good intention, assumptions or speculation. Therefore, there is a need to first understand the underlying principles and premises of screening and then discuss current controversies regarding screening for breast, prostate, and lung cancers.

In this paper we intend to discuss the potential financial, legal, and radiation safety implications associated with whole-body CT or PET cancer screening. The patient's body is scanned and images are preserved. Now using PET/CT abnormalities or cancer tissues are detected in the images. The body parts affected by cancer are localized. A classification is then generated regarding the type of cancer and the stages of the cancer are identified. A model is then created which automatically uses machine learning algorithms to do the task of prediction. The complete process is thus automated.

Keywords : Pet/Ct Scan, Convolutional Neural Network, Artificial Neural Network.,

Application_06

Classification and Localization of Eye Diseases using Convolutional Neural Network

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Abstract :

Due to recent advancements in the field of Artificial Intelligence (AI), many recent strategies have been developed to impartially identify diseases using images. Eye-related diseases are one of the commonly occurring diseases in the human body. Many diseases can manifest in the eye such as Diabetic Retinopathy (DR), Glaucoma, Dry eye, Age Related Macular Degeneration (ARMD), Cataract and so on. These diseases can cause severe discomfort in patients' eyes leading to vision loss, blurred vision, highly impacting on the life of patients. Various AI and image processing techniques have been developed to assist ophthalmologists to diagnose the disease precisely. This idea introduces an automated method of classification and localization of ocular diseases namely Diabetic Retinopathy (DR), Glaucoma, Cataract, etc. using Convolutional Neural Network. In this dissertation we would scan the fundus images of the eye and preserve the images. Now by using CNN we would classify or detect the abnormalities in the images. We would then localize the part which is affected by the abnormality. The next step would include to generate a classification of the type of eye disease. We would then identify the stage of the disease. We now create a model which automatically uses artificial intelligence techniques i.e. CNN to do the task of classification and localization of eye diseases.

Keywords : Artificial Neural Network, Convolutional Neural Network, 2D Images.

Application_07

A Novel Online Shopping Environment Using Interactive Qr Code

Rajratan Pandit Banegaon [1] , L.M.R.J. Lobo [2]

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Abstract :

Human life is getting hectic and challenges are making their space for hurdles to get time even to breathe. Even in this fastest lifestyle getting an array of genuine and quality products, in all ranges is a big challenge. This work and implementation ensured basic and new services in aspects of online shopping as an e-commerce site. In almost all expected things like on time customer assistance, a broad selection range, timely delivery and many more.

An ecosystem of QR code-based e-commerce platforms is constructed to carry business efforts. The idea ensures the quality of each product to be showcased on an online store. The most important part of this system is a new common verification and validation authentication protocol with high security and usability. Social relevance is an utmost consideration while designing and implementing this system. The system will take online orders from customers, accepting their payments, providing information about order confirmation, and many more activities getting involved as a functionality of the system from order placing to order delivery to the customer. The system has proved to be more efficient and accurate as compared to the existing manual systems and is comparable to those developed for pharmaceutical applications which are recently developed.

Keywords : Qr Code, Mobile, Smart Phones, Environment,

Application_08

Ai For Healthcare :Medical Chatbot

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Abstract :

To start a good life, health care is very important. But it is very difficult to consult the doctor for any health issues. The proposed idea is to create a health care Chabot using Natural Language Processing technique. It is the part of Artificial Intelligence that can diagnose the disease and provide basic details about the disease before consulting a doctor. To reduce the healthcare costs and improve accessibility to medical knowledge the Healthcare Chabot is built. Some chat bots act as a medical reference book, which helps the patient know more about their disease and helps to improve their health. The user can achieve the benefit of a healthcare Chabot only when it can diagnose all kinds of disease and provide necessary information. The system provides stextorvoice assistance that means the user can use his own convenient language. Bot will provide which type of disease based on the user symptoms and provides doctor details respective to user disease. And provides food suggestions that means which type of food you must take. Thus, people will have an idea about their health and have the right protection.

Keywords : Chabot ,Artificial Intelligence,healthcare, diagnose, disease

Application_09

Lung Cancer Detection using Convolutional Neural Network Algorithm

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Abstract : Lung cancer is one of the most lethal cancer types; thousands of people are infected with this type of cancer, and if they do not discover it in the early stages of the disease, then the chance of survival of the patient will be very poor. For the suggested reasons above and to help in overcoming this terrible, early diagnosis with the assistance of artificial intelligence procedures most needed. Also, it is one of the most common and contributing to deaths among all cancers. Cases of lung cancer are increasing rapidly. There are about 70,000 cases per year in India. Over the past decade, Cancer detection using deep learning models has been a hot topic, especially in medical image classification. It is worth remarking that CNN models are more advanced at diagnosing diseases such as lung cancers because of the higher performance and ability of the CNN. This system presents an approach that utilizes a Convolutional Neural Network (CNN) to classify the tumors found in the lung as malignant or benign. The accuracy obtained by means of CNN is more efficient when compared to the accuracy obtained by the traditional existing systems. This is done by applying the convolutional neural network technique to a data set of lung cancer CT scans.

Keywords : Lung cancer, artificial intelligence,CNN models

Application_29

Natural Language Processing: A Comprehensive Review

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Abstract :

Natural language processing (NLP) is an active field of research that seeks to develop methods for computers to analyze and understand human language. Natural language processing (NLP) is an interdisciplinary field of computer science, artificial intelligence, and linguistics concerned with the interactions between computers and human language. By utilizing natural language processing, businesses, organizations, and professionals can better understand the meaning behind the text, identify trends in large datasets, and enable automated decision-making. This paper reviews the current state of the field of NLP, its current areas of research, and its potential applications. It also provides a discussion with the help of a case study and also discusses the challenges associated with NLP and our methods to address these challenges. Finally, it looks at the implications of NLP for the future.

Keywords : NLP, Artificial Intelligence, Reinforcement Learning, Machine Translation, Language Generation, Language Understanding

Application_30

Health Care Recommendation System

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Abstract :

Disease Prediction using Machine Learning is the system that is used to predict the diseases from the symptoms which are given by the patients or any user. The system processes the symptoms provided by the user as input and gives the output as predicted disease. K-means algorithm is used in the clustering of the disease which is an unsupervised machine learning algorithm. The probability of the disease is calculated by the Naive Bayes algorithm. With an increase in biomedical and healthcare data, accurate analysis of medical data benefits early disease detection and patient care. This proposed system will not only predict the diseases but also recommend the appropriate doctors based on a particular disease. The list of doctors' datasets will be used for both the symptoms checking and prediction of diseases.

Keywords: Disease Prediction, Healthcare Data, Doctor, Patient, K-Means, Cluster.

Application_31

Questioning Ball (QBall)

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Abstract :

The questioning ball is a toy designed to stimulate curiosity and encourage conversation by presenting random questions for the user to answer. This abstract explores the concept and features of the questioning ball, including its design, construction, and use. The questioning ball can be used by individuals or groups, in educational or recreational settings, and is designed to spark creativity, critical thinking, and communication skills. This also examines the potential benefits of using the questioning ball, including enhanced socialization, improved cognitive function, and increased confidence in public speaking. It provides a description of the design and implementation of the questioning ball, as well as the evaluation of its effectiveness through a pilot study. The results of the pilot study indicate that the questioning ball has a positive impact on students' engagement and participation in the classroom.Overall, the questioning ball is a fun and interactive tool for anyone looking to engage in meaningful conversations and expand their knowledge and understanding of the world around them.

Keywords: Questioning Ball,Speech Recognition,Raspberry Pi

Application_32

Heart Disease Prediction using Machine Learning

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Abstract :

In this paper, machine learning methods and Python programming are used to study the prediction of Heart disease. Heart disease has become a prevalent and deadly condition over the years due to fat suppression. The overpressure in the human body causes this disease to develop. Using a number of characteristics from the dataset, we can forecast cardiac illness. To assess patient performance, we looked at a dataset with 1000 different data values over 12 important parameters. The main goal of this study is to develop algorithms that calculate whether or not a person has heart disease in order to increase the accuracy of diagnosing heart disease.

Keywords: Machine Learning, Prediction, Heart Disease, Symptoms.

Application_33

IoT Based Remote Health Monitoring System

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Abstract :

This paper represents the system for monitoring the patient’s body 24/7 by using IoT. Nowadays, the patient health monitoring system is getting much more popular with researchers and patient guardians. This system has the capability to monitor physiological parameters from the patient’s body at every second. Smart and cost effective healthcare has been in increasing demand to meet the needs of growing human population and medical expenses. There is an urgent need to develop an effective remote health monitoring system that can detect abnormalities of health conditions in time and provide immediate treatment. Recent advances in mobile technology and cloud computing have inspired numerous designs of cloud-based health care services and devices. In this project, an Smart Health Monitoring System (HMS) is built using an IoT platform. The system is responsible for collecting pulse oxygen level, body temperature, ECG from the patient’s body along with the room temperature and sending this data into the IoT Cloud platform by using WIFI-Module and thus the health condition of the patient is stored in the cloud. It enables the medical specialist or Authorized person to continuously monitor a patient's health condition, using data stored on the cloud server.

Keywords : Iot, Remote Health Monitoring, Cloud, Smart Health Monitoring System, Medical Service.

Application_34

Book My Event

A Cross Platform Application for Booking an Event Hall

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Abstract :

To celebrate any kind of event one has to physically visit the event hall and check the condition of the event hall. The event hall is available on the day the user wants to book. To do so, the user has to spend a lot of his time. To tackle this, we are developing an application. This application allows users to check the various Event Halls and their availability with respect to dates. This application will provide the platform between users and event halls owners to book and manage halls. It consists of a list of various Event Halls, which shows the availability according to the date. The application is easy to use.

Keywords : Event Booking, Event Hall

Application_35

Speech Emotion Recognition: An Art of Detecting Emotions through Machines

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Abstract :

Inside the method of education machines assume like human brains, expertise human feelings through his/her speech performs an essential role in Human-pc interaction (HCI). This manner includes numerous steps inclusive of records series, pre-processing, function extraction, characteristic choice, normalization, constructing a classification version with the pleasant accuracy, and for that reason detecting an appropriate emotion of the speaker. This paper covers the discussion over the paintings completed in this field of Speech Emotion recognition (SER) in conjunction with the failed trials and fashions designed to get the appropriate accuracy.

Keywords : Speech, Emotion, Popularity, Convolution, Neural, Network, Mel, Frequency, Coefficients.

Application_36

YOLOv7: A General Framework of Real-Time Multi-Objects Detection For Advanced Driving

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Abstract :

Due to the development of autonomous vehicles, intelligent video surveillance, facial recognition, and numerous people counting applications, there is a great demand for quick and precise object detection systems. These methods locate each object by drawing a bounding box around it in addition to identifying and classifying every object in an image or video. One of the fundamental requirements for autonomous vehicles and most of the Advanced Driving Assistance Systems (ADAS) is the ability to recognize and interpret all static and dynamic objects around a vehicle under varying driving and weather circumstances. Convolutional neural network (CNN) technology has the ability to allow safe ADAS in modern vehicles. This paper analyses the recent deep learning-based object detection methods, challenges and presents an improved general framework for real time multi-object detections in ADAS based on YLOv7 and CNN. We compared the YOLOv7 approach with earlier versions in the YOLO family and it is found that YOLOv7 outperforms all earlier real-time object detectors in terms of accuracy and speed in the range of 5 FPS to 160 FPS.

Keywords : Object Detection, Classification, , Deep Learning,Cnn, Datasets, Training, Adas, Testing, Evaluation

Application_37

Road Traffic Sign Detection and Recognition: A Survey

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Abstract :

Traffic Road traffic Sign Recognition is a branch of applied computer vision research that deals with the automatic detection and classification of traffic signs in images of traffic scenes. The purpose of this review paper is to look after the various classification techniques that can be used to build a system that recognizes road traffic signs in images. The primary goal is to study literature available on the road traffic signs detection and recognition that can identify various types of road signs from static digital images in a reasonable amount of time. In this review paper, we will look at various learning systems for classification that is based on prior knowledge.

Keywords : Traffic Signs, Detection, Recognition

Application_38

Applications Of Artificial Intelligence In Automatic Traffic Count Survey Using Image Processing Technique

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Abstract :

Effective traffic data collection is the cornerstone of the successful planning, maintenance, and control of any modern transport system. Traffic surveys form a major part of the traffic engineer's work and are of fundamental importance since most design problems require detailed knowledge of the operating characteristics of existing and future traffic movements. Accurate data on traffic count for the geometric design of the pavements have become an essential need, as the manual vehicle count survey is associated with various errors. To avoid human errors in the traffic count surveys and save time, an effort is made to computerize the process by detecting the number plates of the vehicles, and input data is then processed to categorize the vehicles.

Keywords : Application Programming Interface (Api), Artificial Intelligence (Ai), Machine Learning (Ml). Modified National Institute Of Standards And Technology Dataset (Mnist).

Application_39

Artificial Intelligence Model for Prediction of Compressive Strength of Concrete

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Abstract :

Various machine learning techniques, which fall under the category of artificial intelligence, are investigated in this work, and an algorithm (model) is developed to forecast the compressive strength of concrete at 7 and 28 days. The results of 180 distinct mixtures with 540 specimens are gathered. The information used in the machine learning model is organized into nine input parameters: cement, fine aggregate, coarse aggregate, water, admixture, compaction factor, w/c ratio, slump, age, and an output parameter: concrete ;s compressive strength. Concrete 7 th and 28 th day compressive strength predictions are based on training and testing results

Keywords : Prediction Of Compressive Strength, Artificial Intelligence, Machine Learning Algorithm.

Theme 3

Research

INDEX

Research_01

Review on Underwater Image Enhancement Using CNN

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Abstract :

Images captured in the underwater environment still suffer from color distortion, loss of detail and contrast reduction due to environmental scattering. The proposed work introduces an improvement approach to enhance the visual quality of underwater imagery. However, images directly captured in the marine environment are still severely degraded, such as undesirable staining, contrast reduction, and detail loss due to light scattering and absorption, that seriously restrict the acquisition of available information from the image. Therefore, acquisition of clear and accurate images is an important prerequisite to help scientists understand the underwater environment. The CNN algorithms are used in various applications such as object detection, tracking, recognition, and surveillance also in navigation. The most critical task here is to separate the water component from the other part. For that purpose, we are proposing an efficient algorithm to remove water from a color image.

Keywords : Convolutional Neural Network, Feature Extraction, Image Classification.

Research_02

An Exploratory Study Investigating The Factors Influencing Career Choice Decision-Making Of Engineering Students: A Data Mining Approach

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Abstract :

Choice decision-making is a complex process influenced by several factors, including personal, societal, and educational factors. This exploratory study investigates the factors influencing career choice decision-making of engineering students using a data mining approach. The study uses the WEKA software and J48 algorithm to analyze data collected from 241 engineering students through a questionnaire survey. The study identifies three main factors influencing career choice decision-making, namely, lack of awareness, availability of information, and cost. The study findings have significant implications for career guidance and counseling programs for engineering students. By understanding the factors that influence career choice decision-making, career guidance and counseling programs can provide tailored support to engineering students, helping them make informed and confident career decisions.

Keywords : Data Mining ,Career Choice Decision Making,

Effect of Artificial Intelligence on Industry 4.0

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Abstract :

This research paper explores the impact of artificial intelligence (AI) on Industry 4.0, the fourth industrial revolution that is characterized by the integration of advanced digital technologies into manufacturing processes. The paper includes a literature review of the role of Industry 4.0, the effect of AI on Industry 4.0, and the benefits to society of AI in Industry 4.0. The research finds that AI is enabling companies to create more efficient, flexible, and responsive manufacturing processes, leading to increased productivity, improved innovation, job creation, enhanced safety, and increased customization. However, there are also challenges associated with the implementation of AI in Industry 4.0, such as the need for skilled workers and the risk of jo b displacement. The paper concludes with recommendations for future research on developing AI applications that can enhance manufacturing processes further while also considering the ethical and social implications of AI.

Keywords: Artificial intelligence, AI, Industry 4.0, fourth industrial revolution, digital technologies, manufacturing processes, efficiency, flexibility, responsiveness, productivity, innovation, job creation, safety, customization, challenges, skilled workers, job displacement, ethical implications, social implications, future research.

Research_04

Approach towardsEnriching Data Quality for Enhanced Decision Making

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Abstract :

Quality of the data is an important aspect in Machine Learning while training a Model for Decision Making. In order to minimize the space, time and algorithm complexity data quality play's prime role. To improve the data quality, pre-processing techniques are implemented on datasets. These pre-processing techniques clean the data obtained from the source and enrich the quality. This paper surveys the factors affecting the data quality and the methods used to reduce the effects of these factors. Data quality enhancement is explained by applying pre-processing techniques on sample datasets. The applications of all the included data cleaning methods are mentioned.

Keywords : Data Quality, Data Pre-Processing,Incomplete Data, Inconsistent Data, Redundant Data, Noise, Outlier, Interquartile Range, Machine Learning, Decision Making

Research_05

Concrete Mix Design by Specific Method of Computerized Grading Kurve Verified by Artificial Intelligence Model

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Abstract :

This paper presents concrete mix design by conventional and computerized method. Incognizance to percentage passing of aggregate and grading of aggregate is observed in case of conventional methods. Based on the desired slump and required water content the calculation starts in conventional method. The important parameter in mix proportioning is fine aggregate and coarse aggregate content that is also determined irrespective of grading of aggregates. A method based on grading of aggregates titled as ‘Specific Method of Computerized Grading Kurve’ has been developed as a modern technique. The automation of ‘SMCGK’ technique has been done by Graphical User Interface based on MATLAB. The mix proportions have been calculated using conventional method and SMCGK technique for M25 grade of concrete. Furthermore, the advanced technology of Artificial Intelligence is applied for getting the predictions of compressive strength of concrete from determined proportions. The decision-making about use of appropriate method of concrete mix design is based on predicted results of the Artificial Intelligence Model. A new computerized technique of concrete mix design developed by MATLAB based Graphical User Interface and implemented with Artificial Intelligence is presented in this paper.

Keywords : Concrete Mix Design, Matlab, Gui, Smcgk Technique, Ai Model

Research_06

Augmented Reality:Redefining Walls

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Abstract :

Augmented Reality (AR) Redefined Walls is a distinguished advancement in terms of redesigning walls. This work is analyzing color shades and suitable artifacts. Via this analysis, trends are extracted and novel insights into promising domains are eventually concluded, from both research and commercial development perspectives. A partial solution is to develop a redefined wall that takes advantage of AR, and design the wall.

Keywords : Ar, Wall Designing, Redesigning Wall

Research_07

The Application of Artificial Intelligence for English Teachers

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Abstract :

The birth and development of science and technology is to better serve mankind. The development of artificial intelligence technology is becoming more and more mature, especially in the application of English teaching, which promotes the reform, development and modernization of English teaching. This paper analyzes the connotation of AI technology and the application of artificial intelligence technology in English listening, writing, translation and oral teaching.

Keywords : Artificial Intelligence, English Teaching, Application.

Research_08

Visualization & Analysis of Geotagged Assets for Project Assessment

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Abstract :

Keeping the track of progress of projects especially at rural or remote areas by the Authorities. At remote sites, the adaptation of technologies is frequently difficult due to a lack of infrastructure and often harsh and dynamic environments. On the other hand, visual inspection by experts usually allows a quick assessment of a project's state. Geotagged data is one of the solutions to check the progress remotely. These geotagged data can be used to visualize and analyze the progress of the sites using big data analysis. Hence there is a need for a digital platform for visualization and analysis & of the progress of the project. This paper proposes the construction progress estimation model based on geotagged data. The process involves capturing the geotagged images using a mobile application. Further it will focus on the visualization of images to check the progress of ongoing projects. Our focus is to standardize the process of construction project tracking at one click and reduce time, efforts and in turn cost.

Keywords : Visualization, Analysis , Progress Tracking, Geotagged Data, Image Difference

Research_09

Failure Prediction Of Rolling Contact Bearings Using Artificial Neural Networks

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Abstract : Bearings are one of the most used elements in the mechanical industry. The main function of rolling element bearing or ball bearing is to allow motion between two rotating parts, with minimum friction. Bearings are an essential component in most applications like motors, gear boxes, vehicle axles, machine tool spindles and many more. As far as mechanical equipment are concerned the most sensitive and common unit of failure are the Rolling Contact bearings. As a wide research area, the prognostics of bearing failure has scope, that helps to have high reliability, reduction of unwanted costs etc. For the safe and optimized operations through lifetime the precise evaluation of Remaining Useful Life is required. To supervise this fundamental component and planning repair work, a novel intelligent method based on the Neural Network approach is proposed. Firstly, the dataset used as input is collected by the Centre for Intelligent Maintenance Systems (IMS). Secondly, the data is used as input data for the deep learning models. The performance of the proposed method is verified by four bearing data sets collected from an experimental setup called "PRONOSTIA". Then, a deep learning-based model is developed to predict the failure of the bearing to determine the life of the bearing. The main objective of the model is to predict the failure with a high accuracy beneficial to bearing life prediction.

Keywords : Roller Bearings; Machine Learning (ML); Deep Learning; Neural Networks; Artificial Intelligence (AI).

Android Malware Detection using Machine Learning

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Abstract :

Mobile phones play a crucial role in today's society given the prevalence of their use in routine activities, from basic ones like alarm clocks to sensitive ones like banking. These gadgets are among the top targets for hackers due to the sensitive and important information they hold. Android-based phones predominate in the phone market. Because of the widespread distribution of malware, Android's open source nature has also given rise to a number of security concerns. For detecting Android malware, multiple classification techniques (individual and ensemble) have been used.In this paper, we propose a Android malware detection system, which uses 5 different types of classifiers for classifying Android apps as benign or malicious. With the proposed method, a comparative assessment is implemented on the performance of popular individual classifiers; experiments on two datasets of malicious and benign Android applications are conducted as well.

Keywords : Android Malware detection, Svm, Dt, Lr,Gb, Random Forest, Machine Learning.

Research_53

Literature Review of portable oil Extraction Machine using Artificial Intelligence machine learning

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Abstract : This paper aims at the design and Development of Seed oil extraction machines. The machine is portable which makes it light in weight and can be used for small scale industries and for Household and Pure oil extraction. Since this machine requires less space to move, it can be used in a more versatile manner as compared to heavy powered machines that are mounted on heavy and large industries. This machine can be efficient and easy to operate and maintain. The oil extracted has various applications in food, medicine, and industry. This paper is also aimed at a Literature Sur-vey on various mechanisms used by top Universities professors According to the Need.

Keywords : Seed Oil Extraction Machines ,Oil Extract , Mechanisms

Research_54

Accuracy Calculation For Body Movement Using Pose Net

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Abstract :

Observing a person&amp’s movement patterns during exercise is an important strategy for personal trainers and exercise physiologists to be able to give accurate feedback on exercise correction during a session. In this project we will be calculating a score, which helps improve the body movements. Pose net is a real time pose detection technique with which we can detect human poses in images or video. Getting 17 key points from pose net and then analyzing them to get the results. Analysis will be shown to the user for understanding which body parts are not in accordance with standard body movements. This work aims to provide a reliable and efficient way to estimate the pose accuracy between two videos, which can be useful in various applications where pose estimation is important.

Keywords : Pose Net, Dynamic Time Warping, Euclidean Matching

Research_55

A review on applications of artificial intelligence and machine learning for the hip and knee surgeon

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Abstract :

Hip and knee replacement surgeries are common procedures for patients with joint degeneration or damage. However, these surgeries require a high level of expertise and precision from the surgeon, and there is always a risk of complications. Artificial intelligence (AI) and machine learning (ML) tech- nologies have been increasingly utilized in various medical fields, including orthopedics. This review paper aims to explore the current applications of AI and ML in hip and knee surgery and their potential benefits.

Keywords: Artificial Intelligence; Machine Learning; Hip Joint Replacement; Knee Joint Surgery; Preoperative Planning

Research_56

Use of Artificial Intelligence for Decision Making

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Abstract

This paper focuses on generally how AI is used everywhere for taking decisions to solve every critical problem, from Govt to people for ease of access for uniqueness in work systems. How AI is used in various ways by using various relevant and appropriate tools and applications in communication, education, art, literature, painting, construction, manufacturing, automation, and housing in general. Artificial Intelligence is a recurrent theme in science fiction. AI in science fiction is a model of a post human situation. Cyborgs have become the most well known part of science fiction literature and other media. This paper also highlights the portrayal of artificial intelligence for creating platforms of data hacking of human beings #39; livesusing virtual reality in science fiction. The depiction of artificial intelligence (machine learning) used to hack the human senses, (cognitive science) brain. The interaction between human beings and technology, highlighting the power of digital in a futuristic, dystopian society. Mechanical organisms have been often conforming to idealistic standards by using such cyborgs to detach the oppressive structures. Because all humans use in some way technology to augment their cognitive processes like naturally born cyborgs. If mind and body are considered one then much of humanity is aided with technology in every way which hybridizes humans with technology.

Keywords— Artificial, intelligence, fiction, simulation,reality, cyborg, cyberpunk, media, chatbot, creativity, human

Research_15

Empowering Smart Cities with IoT and Artificial Intelligence for Better Decision Making

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Abstract

"Empowering Smart Cities with IoT and Artificial Intelligence for Better Decision Making" is a comprehensive report that examines the current state of Internet of Things (IoT) technology, including its history and emerging areas, as well as its limitations and areas that need improvement. The report focuses on the potential applications of IoT in building smart cities, exploring how IoT can empower decision making in urban planning, transportation, energy management, and more, with the help of Artificial Intelligence (AI) techniques. The report also delves into the challenges facing the implementation of IoT in smart cities and concludes with recommendations for future development, emphasizing the importance of collaboration, standardization, and increased investment in research and development. Ultimately, this report aims to contribute to the ongoing advancement of IoT and AI for the betterment of our cities and communities.

Keywords-Internet Of Things (Iot),Artificial Intelligence (Ai),Planning, Transportation, Energy Management

Theme 4

Hardware Relevant

INDEX

Hardware Relevant_01

Dyslexi Learning Device

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Abstract

For children with disabilities in minor or major forms in any of the physical organs, or for mentally unstable children, we need to consider and provide them with a supportive growing environment. Dyslexia is most commonly associated with trouble in learning to read. It affects a child's ability to recognize and manipulate the sounds in language, a learning disorder with impairments in reading and spelling acquisition. To overcome this problem, we built a learning module system for the children. Our main motto is related to their education, which aims to improve reading and spelling acquisition. Each little one chooses an activity that interests them and takes learning in their own space. To promote practical life skills and academic ability, in this learning module, a Controller based puzzle game is designed for spelling and image recognition and solving basic mathematical operations. Based on the performance of the child, the analysis response can be stored on the cloud and can be used further for decision-making of the condition and level or stages of dyslexia in the child. In this work, simple daily routines and activity patterns for learning are included to help children manage their feelings. This module will help children to be confident, active learners who are proud of what they know and can do.

Keywords—Dyslexia Disorder, Dysgraphia, Dyscalculia, Learning Device, Learning Disorder, Learning Module, Gaming Device, Fun To Learn Device.

Hardware Relevant_02

Automation in Device Control Based on GSM

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Abstract

In this modern world, analog electronics devices are becoming less popular and digital electronics components are becoming more advanced and replacing the former, day by day. Home devices control systems are one such example of a modernized digital world. People are using cellular mobile networks to communicate with each other. GSM modules are one of the basic elements of these phone networks. GSM, which stands for 'Global System of Mobile communication', is also used in many electronics projects among engineering students and also in the industry. GSM Based projects are used to control end devices through a mobile device from remote locations. For example, if a user wants to control any machine from a remote location using a mobile phone, can they do it? The answer to this question is yes. Users can easily turn on and turn off devices using GSM modules and mobile phones. In this project an automation system is built where users can control appliances, using even a simple GSM based phone, by means of a simple SMS from that phone.

Keywords-- Mobile Network; GSM; SMS

Hardware Relevant_03

Solar Jacket

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Abstract

The most significant issue in extremely cold temperatures is hypothermia. A low body temperature affects the brain, causing the person to be unable to think clearly or move well. Soldiers must do their duties in extremely cold climatic conditions where their body temperature lowers rapidly, so they must maintain their body temperature. Regular sweaters and sweatshirts don't provide much warmth, and they are useless in really cold weather. The Solar Jacket can help you keep your body temperature stable. The wearer can change the temperature to his or her preference. For the solar jacket's heating element, we chose carbon fiber heating wire, which can disperse heat evenly and give the body comfortable warmth. The solar jacket can be used even by people having colds and fevers. The solar jacket can also be powered using an external power supply in case there is not enough sunlight. The solar jacket can also be used as a normal jacket for everyday use. Solar Jacket also allows users to carry, connect and charge their portable devices

Keywords : Solar Jacket, Power Supply, Portable Devices

Hardware Relevant_04

Evaluation And Observation Of Thermal Energy Storage System

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Abstract

TES can impact air emissions at building sites by reducing the amount of ozone-depleting CFC and hydro chlorofluorocarbon (HCFC) refrigerants in chillers and the amount of combustion emissions from fuel-fired heating and cooling equipment. TES helps reduce CFC use in two main ways. First, since cooling systems with TES require less chiller capacity than conventional systems, they use fewer or smaller chillers with less refrigerant. Second, using TES can offset the lost cooling capacity that sometimes can occur when existing chillers are converted to more benign refrigerants, making building operators more willing to switch refrigerants Thermal energy storage (TES) generally involves the temporary storage of high- or low-temperature thermal energy for later use. Examples of TES are storage of solar energy for overnight heating, of summer heat for winter use, of winter ice for space cooling in summer, and of heat or cool generated electrically during off-peak hours for use during subsequent peak demand hours. In this regard, TES is in many instances an excellent candidate to offset this mismatch between thermal energy availability and demand

Keywords : Ozone-Depleting Cfc, Thermal Energy Storage (Tes), Chlorofluorocarbon (Hcfc)

Hardware Relevant_05

Estimating airfoil properties using SVM -Regression

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Abstract :

SVMs or Support Vector Machines are one of the most popular and widely used algorithms for dealing with classification problems in machine learning. However, the use of SVMs in regression is not very well documented. In the current study SVM has been used for predicting airfoil properties at intermediate angles using polynomial regression. Support Vector Regression uses the same principle as the SVMs. The basic idea behind SVR is to find the best fit line. In SVR, the best fit line is the hyperplane that has the maximum number of points. The component described in this paper is a part of a larger software that has developed to automate designing and testing of airfoils.

Keywords -Svm, Svr, Airfoil, Cfd, Drag Coefficient, Lift Coefficient

Hardware Relevant_06

48V Electric Vehicle Battery Pack

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Abstract

A revolution in the automobile sector is taking place in the form of Electric Vehicles. They have improved exponentially in the terms of quality and technology in recent times. The main key components which drive Electric Vehicles are motors, battery packs, chassis, and the technology that controls all the parameters. In this paper, we are going to study battery packs and the technology that is used to run them seamlessly. And about the battery packs, they have main components such as lithium-ion cells, BMS (battery management system), connectors, connectivity system, cooling system, etc. There are some external factors also that affect the battery performance like environmental conditions and temperature, which are very important while operating an Electric Vehicle, and this can be controlled or regulated by the cooling system present in the battery pack. Advancements in technology and connectivity have opened doors for every industry due to which everything is connected to the web and becoming part of the Internet of Things (IoT). We have cloud connectivity to battery packs also, so that we can take data in real-time from the battery pack which helps us to make decisions and perform some tasks also.

Keywords : Lithium-Ion Battery, Battery Pack Model, Battery, Cell Model, State Of Charge, Cooling, Bms, Iot, Cloud.

Hardware Relevant_07

Prediction of tool failure in CNC Machine using Machine Learning

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Abstract

Machine learning (ML) is a technology which is a branch of Artificial intelligence (AI) focused on learning based on Data, Algorithms, Graphs, and different statistical models in order to adopt the technique humans learn and hence increase in the accuracy by the time. For example, every time you search on an e-commerce website and get a wrong result based on the feedback provided form us the algorithm keeps on improving. In the era where there is a 4 th revolution in the industries there is a need for more automation and more efficient way of production in industries. One of the ways we can achieve it is by predicting the maintenance before the failure occurs to bring this into reality. Machine learning and artificial intelligence methods have been proven efficient. In concern with this problem, we have come up with a solution by building some machine learning algorithm which can provide us with some accurate predictions which will be helpful in saving a lot of time in the industry. It can also be used for detection to figure out that something is going to happen or how long a part will work before there is a strong chance that it is going to fail. It is also used to understand the root cause

Keywords—Machine Learning, Artificial Intelligence, Data, Algorithm, Predictive, Maintenance

Hardware Relevant_08

Literature Review of tire life for automobiles using AI Technology

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Abstract

Automobiles have evolved into an indispensable form of transportation with the expansion of the economy. Automobile tires are important components, and tire failure contributes to numerous traffic collisions. As a result, one of the crucial elements affecting a vehicle & #39; s safety is anticipating tire life. An image processing and machine learning-based strategy for predicting tire life is presented in this research. As a starting point, we first create a database of unique images. In engineering practice, there are typically only a few numbers of sample image libraries, hence we suggest a novel image feature extraction and expression method that has great performance for a limited sample collection. Using the gray- gradient co-occurrence matrix (GGCM) and the Gauss-Markov random field (GMRF), we extract the texture features from the tire image, and then classify the retrieved features using the K-nearest neighbor (KNN) classifier. After that, we run trials to determine how long car tires will last. As evaluation criteria, the confusion matrix and mean average precision (MAP) are used to estimate the experimental outcomes. Lastly, we evaluate the efficiency and precision of the suggested approach for predicting tire life. The discovered results are anticipated to be used for real-time tire life prediction, hence lowering traffic incidents involving tires

Keyword- Grey- Gradient Co-Occurrence Matrix, K-Nearest Neighbour (Knn), Mean Average Precision (Map)

Application of AI in Ambient air quality monitoring in solapur city

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Abstract :

In order to better understand the applications of big data and AI technology in environmental protection monitoring, an analysis has been conducted based on atmospheric science. Furthermore, a combined model of air quality forecasting that employs machine learning has been suggested as a solution to current air quality monitoring challenges concerning environmental protection. The proposed model consists of ice emdan-woa-elm; an improved complete ensemble empirical mode decomposition with adaptive noise whale optimization algorithm-extreme learning machine. by leveraging deep learning, a time-space-type-meteorology (tstm) model was established to predict air quality. To test the accuracy of this model, an experiment was conducted and the results revealed that it outperformed a single ai model in forecasting air quality. Specifically, the five evaluation index values of iceemdan-woa-elm were 18.621, 21.986, 0.031, 0.018 and 1.671--these values were higher than those from any other models used in comparison. In regards to air quality forecasting, it is evident that a single ai model cannot keep up with current requirements. However, examining results from the tstm model showed an impressive accuracy and average score of nearly 1.00 with a maximum of 1.00 when predicting heavily polluted weather conditions. Furthermore, its performance decreased slightly as step size increased yet stayed above 0.86, showing potential for environmental protection using big data combined with the ice emdan-woa-elm model.

Keyword: Machine Learning, Iceman-Woa-Elm,Artificial Intelligence

Prediction of Northeast Monsoon Rainfall over Peninsular India using Climate Indices by Genetic Programming Approach

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Abstract

The Monsoon Rainfall And The Resulting Streamflow In Indian Rivers Have A Remarkable Impact On The Socio-Economic Aspects Of India As The Agricultural Sector Majorly Depends Upon Monsoon Rainfall. It Is Crucially Important To Understand The Rainfall Variability, Its Connection With Other Climatic Parameters And Its Prediction As The Indian Economy Is An Agrarian Economy. 'Northeast Monsoon' (Nem) Is One More Form Of Monsoon Activity Experienced By The Peninsular Indian Region. The Nem Contributes Anywhere Between 30% And 60% Of The Annual Mean In Meteorological Subdivisions Of Tamil Nadu, Kerala, Coastal Andhra Pradesh Rayalaseema. Nem Season Is Important From The Point Of Stream Flow, Reservoir Yields, Water Resources Availability For Agricultural And Other Purposes. Despite Having Significant Agricultural And Economic Importance, The Nem Has Been Considerably Understudied As Compared To The Summer Monsoon. Therefore Predicting Nem Rainfall Is Quite Important. A Wide Range Of Rainfall Forecast Techniques Is In Use For Weather Forecasting At Regional And National Levels. The Most Widely Used Techniques For Climate Prediction Are Regression, Multi-Target Regression, Artificial Neural Network, Recurrent Neural Network, Fuzzy Logic, And Genetic Programming. In This Work Genetic Programming Approach Is Adopted Using Discipulus Software For The Prediction Of Nem Rainfall Over Tamil Nadu, India. For The Prediction Of Nem Rainfall Global Climate Indices Like Enso, Equinoo And Olr Were Considered. It Has Been Observed That Genetic Programming Has Given Satisfactory Results. The Value Of Pearson's Correlation Coefficient (R) Between Observed And Predicted Rainfall Was (R=0.80) For Training Phase And (R=0.73) For Testing Phase, Nse For Training Phase (0.63), Testing Phase (0.57) And Rmse For Training Phase (61.69), Testing Phase (55.93) Are Satisfactory.

Keywords : Northeast Monsoon, Rainfall Prediction, Genetic Programming, Peninsular Indian Region.

Hardware Relevant_70

Review paper on use of artificial intelligence in decision making for the maintenance and repairs of civil structures.

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Abstract :

A Powerful Tool In Artificial Intelligence, According To Several Experts, Neural Networks Can Evaluate And Maintain Bridge Constructions And Other Civil Structures With More Than 90% Accuracy. Using Artificial Intelligence Improves Safety On Building Sites. The Majority Of The Research In This Field Has Concentrated On Employing Digital Twins, Machine Learning, And Artificial Intelligence To Track, Improve, And Maintain The Accuracy Of Bridges And Other Civil Constructions. This Study's Major Objective Is To Look At How Artificial Intelligence May Advance The State-Of-The-Art In Data-Driven Maintenance And Repairs Of Bridges And Other Civil Infrastructure..

Keywords*:* Neural Network, Fuzzy Logic, ,Bridges, Digital Twin Technologies, Buildings

AI has been around for more than three decades, nowadays, it has shown an exponential growth in its developments of applications for society, economy, and military. It's involvement is changing so rapidly and it's deployments are highly visible. It has changed the way we live and interact. Artificial Intelligence (AI) therefore has been redefining society. We sometimes do not even anticipate what results it can produce. Technology is clinging to us, actually making us totally dependent in every phase of our life, right from switching on our smartphones to our every day activity. This may need applications for online shopping, intelligent car dashboards and autonomous robots. Forming a basis for many computer learning and complex decision-making processes has made things happen. The availability of powerful tools for processing huge amounts of data has made this field of technology to pick up pace.

AI is used as a tool for developing innovations. We can experiment with it. It shows excellent results in almost all Indian domains, including healthcare, education, agriculture, finance, automobiles, energy, retail, manufacturing, scientific research with stand alone discoveries in place. In India, companies like Walmart, Google, Microsoft, Amazon, Samsung are into AI-based research. Still, a lot of potential to expand in its research is uncovered in this cutting-edge technology AI. Most of our educational, government and private institutes nurture and motivate AI researchers, innovations and start-ups.

Business over the years has evolved in baby steps from local corner shops to the booming platforms for shopping online. These modern techniques make individual lives easier. They streamline business processes, improve user experience, sales forecasting and automate the decision making process to show successful results. Today's business world is solely dependent on AI, Cloud, Big Data technologies of which e-commerce and m-commerce are main applications. In synch with global developments in innovation and automation, India too has brought about a digital transformation over the last two decades. Now, technological developments have gained pace more than what has been predicted; the pandemic played a great role in its quick transformation and adoption.

The Artificial Intelligence for Decision Making Initiative heads to develop Artificial Intelligence (AI) and Machine Learning (ML) expertise and capability in areas of significant importance to the Australian defence and national security community.The Government is pushing the private sector and offers many opportunities through DST, Niti Aayog, IndiaAI and many more, to create innovative technological solutions and fund AI-based start-ups. The start-ups are focussed in the cities like Bengaluru, Hyderabad, Ahmedabad, Mumbai, Delhi for AI-based businesses.

This Two days National level conference was an interdisciplinary one where there were participants from various parts of the country. The participants presented research carried by them having social significance and developing systems of automation in their own domains exhibiting the use of AI for taking decisions to solve problems. The reviewers and session chairs consisted of persons from industry and academia. This proceeding is a collection of all abstracts of the papers submitted for this conference.

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